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**Lyceum of the Philippines University General
Trias, Cavite**

**College of Engineering, Computer Studies, and Architecture
Department of Computer Studies
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THESIS PROJECT CAPSULE

**Approved Title: AI-Driven System for Efficient Scheduling and Performance
Assessment of College Research Presentations in Lyceum of the Philippines
University of Cavite Department of Computer Studies (LPU-C DCS)**

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Program and Section: CS 401

I. GENERAL SUBJECT AREA

Software Development

II. SPECIFIC TOPIC

III. SPECIFICS OF RESEARCH

INTRODUCTION

Nowadays, Artificial Intelligence (AI) has become increasingly refined each day to provide more context and definitions for things we want to understand and learn about. AI also contributes by offering deeper context and definitions for topics we want to explore. AI can assist us in finding various definitions and perspectives for different studies or theses. However, there are aspects that AI cannot fully address, such as creating new and original thesis topics, automatically scheduling, grading systems, listing students, teachers, panelists, and the timing of thesis reviews.

One of the problem of the students of Information System, Tanjungpura University (Ulsa, M. A.-F. A., Sari, R. P., & Febriyanto, F. 2023) has the difficulties in selecting a suitable thesis fields due to a lack of structured and objective guidance, resulting in delays and topic mismatches , and thus they created a black box A decision support system has been developed to facilitate students in determining the field of study for their theses. The system uses the profile-matching method to provide mathematical and accurate recommendations for the thesis's field of study. The system testing employs the black box testing method. A decision support system has been created to help students choose their thesis field of study. It uses a profile-matching approach to offer precise and mathematically based recommendations. The system has been evaluated through black box testing.

Selecting a research topic can be hard, that is relevant and can be a significant challenge for students and researchers. The existing methods often rely on outdated information or lack personalized guidance, resulting in a potentially redundant study and ineffective topic selection.

To address this issue, (Pandit, Dr. H. B., & B. Shah, Dr. D. 2017) proposed Decision Support System (DSS) model aims to enhance the research topic selection process through a combination of decision support systems and web content mining. The DSS model requires students to answer a set of predefined questions using keywords, which are then used to search for relevant research topics across various online platforms. This approach is designed to present students with the most current and relevant research topics, aiding them in choosing a suitable research area. Additionally, supervisors can use the system to explore emerging research trends and access a broad range of links related to specific keywords. The system also includes a feature to search existing research work to prevent unintentional plagiarism.

It is good to find ways to make LPU-Cavite Professor, department head or research professor have easier to manage their Scheduling, that is where our project, "AI-Driven System for Efficient Scheduling and Performance Assessment of College Research Presentations in Lyceum of the Philippines University of Cavite Department of Computer Studies (LPU-C DCS) ," comes in. This will help them lessen the stuff that they do.

Project Context

According to Valentino, V. H et al, (2021) humans can make a human mistake, to which that decision will favor one part, So he proposed a system that will help eliminated, by taking an absolute decisions, that will be not changed through any process so that the graduation decision from a thesis trial that the students do can be with a definite decision and the graduation can be sure if all the requirements had been fulfilled.

According to Pierce, K., et al. (2022) CMS tools have become one of the significant technological developments, offering open, large-scale-web-based courses accessible to anyone with internet, Using CMS students and professors have been benefits them by providing them relevant and useful data.

The thesis topic decision tool will suggest a relevant topic based on current computing trends, helping the students choose quality research projects. The research title acceptance tool will compare the proposed titles with existing studies, ensuring uniqueness and avoiding plagiarism. The scheduling system will help automate the scheduling of thesis defenses, based on the availability of students and panelists, providing them the notifications and rescheduling options if needed.

The content management system will allow the faculty to modify the rubrics, team lists, and the other data, ensuring up-to-date information. The requirement checker tool will track the students' progress and notify them of the missing requirements before their defense. Lastly, the pass recommendation system will analyze the panelists' feedback and scores, offering them an objective recommendation on whether the thesis has been successfully defended. Moreover, the Web will be to ensure System testing, Integration Testing, Unit Testing, Functional Testing, Usability Testing, User Acceptance Testing, Performance Testing and Security Testing

The Project will make use of Generative Pre-trained Transformer (GPT) Based model for the AI-driven-web-based system that will help the professor give better feedback and results for the student defense.

The Support Vector Machine (SVM) will help to suggest a relevant and trending topic based on students' interest and the current computing trends, it will help analyze the

proposed research titles for uniqueness and the relevance of the title, Help the panelist about the feedback and if the student has passed their defense.

The project will help the improve the quality-of-life of Professor, department head or research professor encouraging them to be more focused to do Assessment.

Purpose and Description of the Study

The objective of this study is to create an AI-driven web-based system that enhances scheduling, evaluation and feedback process of research presentations at Lyceum of the Philippines University Cavite (LPU-C), specifically within the Department of Computer Studies (DCS).

The Proposed website will have features such as thesis topic decision tool, research title acceptance tool, scheduling system, content management system, requirement checker tool, and pass recommendation system. The features mentioned are meant to assist professors and students to have better feedback and output for research presentations.

Statement of the Problem

The purpose of this research is to develop a web application called AI-Driven System for Efficient Scheduling and Performance Assessment of College Research Presentations whose purpose is to do away with the following problems:

1. Confusion of use of the existing system:

- Professors find the system challenging, leading to errors in information due to haste, potentially causing disruptions in scheduling and assessment.
2. Inability to track the history of panelists:
 - Missing historical panelist data hampers accuracy and accountability, impeding smooth panelist rotation and record-keeping.
 3. Conflicting Schedule of panelist and defense:
 - Inadequate coordination between panelist meeting and defense times leads to last-minute changes, affecting feedback quality and student schedules.
 4. Inability of the students to give requirements on time:
 - Late submission of student requirements results in penalties, lower grades, and increased workload for both students and faculty, creating inefficiencies and stress in the academic process.

Objectives of the Study

The project will develop AI-Driven System for Efficient Scheduling and Performance Assessment of College Research Presentations, a web application aimed at enhancing the existing system used for college research presentations.

Specifically, the project aims to:

1. Develop an AI-Driven System for Efficient Scheduling and Performance Assessment of College Research Presentations system with the use of the following feature:

- a. thesis topic decision tool
 - Suggests relevant topics based on current trends to help students choose quality capstone/thesis projects.
- b. research title acceptance tool.
 - Analyzes existing studies to assess the uniqueness of a title, avoiding plagiarism or low-quality topics.
- c. scheduling system
 - Aligns the defense schedules of both students and panelists based on availability.
- d. content management system
 - Allows modular modification of rubrics, team lists, title lists, and panelist assignments.
- e. requirement checker tool
 - Helps students and advisers track the completion of necessary documents before the defense.
- f. pass recommendation systems.
 - Provides decision support based on panelists' feedback to determine if the defense was successful.

2. The project will be developed using HTML / CSS and JavaScript for front-end flexibility and a user-friendly interface. PHP and Python will be used for the backend to manage server-side logic and real-time data processing. For database management, MySQL will be utilized to provide scalable and efficient data storage.

3. Evaluate and improve the system that will be developed using:
 - a. System tests: Verify overall functionality and behavior.
 - b. Integration tests: Ensure seamless communication between frontend and backend.
 - c. Unit tests: Confirm testing to ensure individual features work components function correctly.
 - d. Functional tests: Validate Integration testing to verify that different components work together as intended.
 - e. System testing to evaluate the complete and integrated application against specified requirements.
 - f. Usability tests: Evaluate user-friendliness and intuitiveness.
 - g. User acceptance testing (UAT) to ensure the application meets the needs and expectations of the users.
 - h. Performance tests: Assess testing to assess the application's speed, scalability, and stability and responsiveness under various conditions load.
 - i. Security testing to identify and fix vulnerabilities to protect user data and maintain privacy.
4. Evaluate the performance of the system and measure it based on the following criteria, as per ISO 25010 standards for software development.

Scope and Delimitations of the Project

The AI-Driven System for Efficient Scheduling and Performance Assessment of College Research Presentations focuses on improving the process of conducting and presenting the research activities at the LPU-C, particularly within the DCS. This system is however aimed at enhancing the processes of scheduling, assessing and soliciting feedback on college research presentations. Among its notable features are the Thesis Topic Decision Tool that assists students in identifying research areas by recommending some relevant ideas from the current areas of computing and the Research Title Acceptance Tool that scans the proposed titles against previously studied titles to protect them from similar ideas.

A Scheduling System which allows one person to manage research defense schedules for students and panelists by determining their availability, preventing conflicts and providing rescheduling options when needed, was also included in the system. The Content Management System is used to revise rubrics and up-to-date team and panel assignments of faculty members. The constituents are satisfied with such changes and features and other problems such as the Requirement Checker Tool, a tool that seeks to ensure students fulfill their offense requirements before a scheduled defense, educates them on requirements per future deadlines. Lastly, the evaluation of the feedback collected by panelists and the provision of the Pass Recommendation System based on the analysis of the feedback will provide an assessment of the defense undertaken by the student on the research work done.

The system uses artificial intelligence and machine learning such as the GPT for suggestion generation and a SVM for title and feedback analysis. The system aims at

eliminating the risk of human error, potential or actual overlap in scheduling and obtaining useful corrections within a short period to improve the flow of work in students and in teaching staff as well.

The project would only include its focus on the Department of Computer Studies (DCS) at LPU-C, which implies that it is designed only for this department and may not be applicable in other departments without alterations. This system does enhance the system for scheduling and assessing the presentations of the research done but does not include other administrative or academic activities of the university. Furthermore, the main founders and the copyright owners have legal rights over the system and its content because the system is web based which implies that it requires internet connection and suitable devices. Grants for Research Title Acceptance Tool do place some measure for checking uniqueness, however; it does not check for plagiarism in respect of the whole document. Also, in as much as the decisions have been assisted through Artificial Intelligence, there is still the requirement of human resources within the institution to enhance the accuracy and fairness of the grading exercise.

Schedule of Activities

The table below shows the schedule of activities of the proposing system:

Activities	Months					
	1	2	3	4	5	6
Planning Phase	■					
Research and Analysis		■				
Design Phase		■				
Development Phase			■	■	■	
Testing and Iteration				■	■	
Deployment and Evaluation					■	■

Resources

The proponent will consider the following resources:

Reference:	Internet resources, academic literature, research papers, LPU-C regulations and guidelines related to thesis procedures
Hardware:	Devices compatible with common browsers such as Google Chrome, Safari, Mozilla Firefox, Microsoft Edge, Opera
Software:	HTML, CSS, JavaScript, Figma, MySQL, Windows, Android, iOS, GitHub

Algorithm: Support Vector Machine and, Generative Pre-trained Transformer based model.

Expected Output

The expected output of AI-Driven System for Efficient Scheduling and Performance Assessment of College Research Presentations in Lyceum of the Philippines University of Cavite Department of Computer Studies (LPU-C DCS) is the web application and implementation of a web-based system that will effectively help the professors to give better feedback that suits the student's performance during defense.

By utilizing artificial intelligence (AI), the Pass Recommendation tool consists of sentiment analysis that will use the professor's feedback and give results if the student deserves to pass. Another feature is the Scheduling system. Professors, department head or research professors could post a schedule on the site, students could also see their upcoming thesis defense posted on the website. Next feature is the content management system, this feature allows students to filter their respective criteria depending on their track/course. Next is the Requirement Checker Tool, this allows students to track their requirements when it comes to passing chapters for their thesis/capstone. Research Title Acceptance Tool is another feature in which the system will check if the student's title has similarities to current and past thesis/capstone titles, the system will provide feedback if the title is unique or plagiarized. Lastly is the Thesis Decision Tool, the system will find current trends when it comes to computing solutions and will make suggestions or ideas for the thesis of the students.

The site will feature a user-friendly user interface that will suit the professor and student's needs. The system will be scaled depending on semesters and will get its data from available sources in the school.

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